CASE 3

Ms. Sophie Bennett — 34F · RRMS | Post-relapse (optic neuritis) · Fatigue · Leg heaviness >200 m · Bilateral foot paraesthesia · Balance impairment when fatigued · Fear of exercise-triggered relapse

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- • Primary central MS fatigue
- • Bilateral lower limb sensory impairment (paraesthesia — feet)
- • Balance impairment, worse with fatigue and heat
- • Functional lower limb weakness / leg heaviness on exertion
- • Likely residual visual deficit from optic neuritis
- • Walking limited to <200 m before symptom onset
- • Reduced confidence and safety with community ambulation
- • Fall risk (balance compromised under fatigue loading)
- • Reduced community participation (outings, social activities)

■■ RED FLAGS

- ■ New or worsening bladder / bowel dysfunction
- Significance:
- Urgent MS nurse / neurologist notification. Continence physiotherapy referral. Withhold intensive exercise until relapse ruled out.
- ■ New neurological deficit within the past 4–6 weeks (possible active relapse)
- A relapse is defined as new or worsening neurological symptoms lasting >24 hours in the absence of fever/infection. Exercise during an active relapse can worsen symptoms and delays recovery.
- Refer back to neurologist urgently. Suspend formal exercise programme until relapse confirmed resolved (typically ≥4 weeks post-relapse onset).
- ■ Lhermitte's sign (new onset)
- An electric shock sensation down the spine on neck flexion — indicates active cervical cord demyelination. New onset suggests an evolving lesion or early relapse.

■ PATHOPHYSIOLOGY (KEY POINTS)

- Myelin forms the insulating sheath enabling rapid saltatory conduction between nodes of Ranvier; its destruction dramatically slows or completely blocks action potential propagation.
- Clinically, this produces a discrete neurological deficit corresponding to the anatomical location of the active plaque — Sophie's optic neuritis reflects inflammation at the retrobulbar optic nerve, causing painful visual loss.
- During remission, inflammation partially resolves and oligodendrocyte precursor cells attempt remyelination — which accounts for partial recovery of function.
- However, remyelinated axons have shorter, thinner internodes and conduct more slowly; they are also more metabolically vulnerable.
- Sophie's bilateral foot paraesthesia reflects dorsal column demyelination at spinal cord level, disrupting proprioceptive and fine-touch afferent input to the brain.
- Her central fatigue arises because the brain must work at significantly higher cortical effort to drive signals across the inefficient demyelinated network — this is metabolically exhausting and cannot be reversed by willpower alone.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations

Tone: Assess bilateral lower limb tone at rest and through passive range. In post-relapse RRMS, mild spasticity (UMN pattern) may be present. Check for ankle clonus on rapid dorsiflexion. Coordination: Heel-shin test (cerebellar/proprioceptive pathway integrity); finger-nose test if upper limb symptoms present. Look for intention tremor, dysmetria, or dysdiadochokinesia. Posture:

Berg Balance Scale (BBS)

Functional Balance What it measures: Static and dynamic balance across 14 functional tasks, scored 0–4 each (total 0–56). A score below 45 indicates significant fall risk in the community-dwelling MS population. How to perform: Set up a standard chair, a small step, and a 5-metre clear space.

■ TREATMENT PLAN

- 2. Can you describe the leg heaviness when you walk — at what point does it come on, does it affect both legs equally, and what does the walking look like when it gets bad?
- 3. Describe the pins and needles in your feet — constant or intermittent, does it affect your confidence or balance on different surfaces?
- 4. Have you had any falls or near-falls in the last three months — and does your balance feel different in the dark, or on uneven ground?
- 5. You mentioned being terrified exercise might cause a relapse — can you tell me more about where that belief came from?
- 6. How is your vision since the optic neuritis — any residual blurring, colour change, or depth perception problems?
- Test 2: Romberg Test and Tandem Romberg
- Sensory Integration
- What it measures:

• Patient stands with feet together, arms by sides. Observe sway with eyes open for 30 seconds — note amplitude and direction.
• Ask patient to close eyes; maintain the same position for 30 seconds. Remain in close standby to catch.
• Positive standard Romberg = marked increase in sway or loss of balance with eyes closed.
• Advance to tandem Romberg: one foot directly in front of the other (heel-toe), arms folded. Eyes open, then eyes closed.